Matrix 2 tutorial 3

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1 Introduction

Tutorial Sheet 3
Matrix Algebra & Advanced Calculus II

Part I: Quadratic Forms and Sylvester Criterion

Problem 1. Consider the quadratic form

$$Q(x, y, z) = 4x^2 + 2y^2 + 3z^2 + 4xy - 2xz + 2yz.$$

1. Write the associated symmetric matrix.
2. Use the Lagrange method to reduce Q to diagonal form.

Problem 2. Consider

$$Q(x, y, z) = x^2 + 2y^2 + kz^2 + 2xy + 2yz.$$

1. Write the matrix of Q.
2. Use Sylvester's criterion to determine for which k the form is:
   - positive definite,
   - negative definite,
   - indefinite.

Part II: Gradient and Level Sets

Problem 3. Let

$$f(x, y) = x^3 - 3xy^2.$$

1. Compute $\nabla f(x, y)$.
2. Find critical points.

3. Determine direction of maximal increase at $(1, 1)$.

Problem 4. Consider

$$f(x, y) = x^2 + y^2 - 4x.$$

1. Describe level sets geometrically.
2. For which c is the level set nonempty?
3. Sketch level sets.
4. Compute gradient and interpret geometrically.

Part III: Hessian Matrix and Local Extrema

Problem 5. Let

$$f(x, y) = x^4 + y^4 - 4xy.$$

1. Find critical points.
2. Compute Hessian matrix.
3. Classify critical points using second derivative test.
4. Discuss the result.

Problem 6. Let

$$f(x, y) = x^4 + y^4.$$

1. Show $(0, 0)$ is critical.
2. Compute Hessian at $(0, 0)$.
3. Explain why test is inconclusive.
4. Determine nature of critical point.

Part IV: Optimization with Constraints

Problem 7. Find extrema of

$$f(x, y) = x^2 + y^2$$

subject to

$$x + 2y = 6.$$

Problem 8. Find extrema of

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to

$$x + y + z = 3.$$

Part V: Min/Max on a Domain

Problem 9. Find global extrema of

$$f(x, y) = x^2 + y^2 - 2x$$

on

$$x^2 + y^2 \leq 4.$$

Problem 10. Find absolute extrema of

$$f(x, y) = x^2 - 4x + y^2$$

on

$$0 \leq x \leq 3, \quad -1 \leq y \leq 2.$$